

# Credit Card Default Auto-ML Report — ensemble (sub)

 2026-05-28 22:04:35 UTC

모형 ensemble

roc\_auc = 0.7663

BEST(catboost) = 0.7796

Δ vs best = -0.0133

## 1 점수 요약 (본 모형 vs best)

| metric    | ensemble (현재) |        | catboost (BEST) |        | Δ(Test) |
|-----------|---------------|--------|-----------------|--------|---------|
|           | Train         | Test   | Train           | Test   |         |
| roc_auc   | 0.7902        | 0.7663 | 0.8099          | 0.7796 | -0.0133 |
| ks        | 0.4389        | 0.4168 | 0.4660          | 0.4362 | -0.0195 |
| lift      | 3.2341        | 3.0595 | 3.4037          | 3.1274 | -0.0678 |
| accuracy  | 0.8248        | 0.8153 | 0.8308          | 0.8177 | -0.0023 |
| precision | 0.7058        | 0.6585 | 0.7133          | 0.6589 | -0.0005 |
| recall    | 0.3569        | 0.3429 | 0.3931          | 0.3640 | -0.0211 |
| f1        | 0.4741        | 0.4509 | 0.5069          | 0.4689 | -0.0180 |
| pr_auc    | 0.5809        | 0.5377 | 0.6194          | 0.5588 | -0.0211 |

## 2 Feature Importance — ensemble

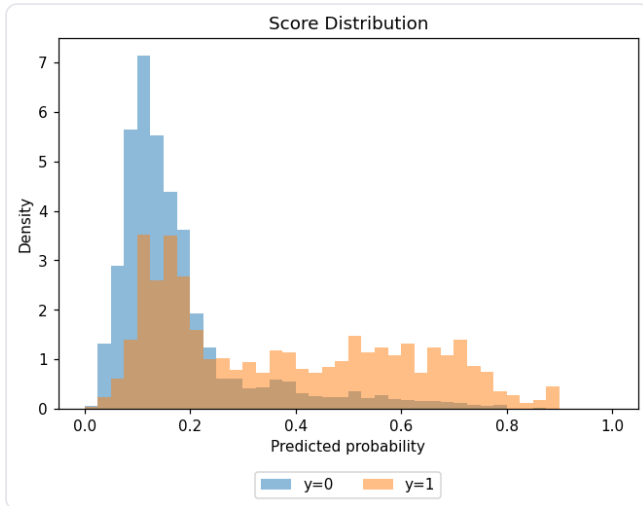
relative · top 19 · 전체 분포는 feature\_importance.csv

| Rank | Feature   | Importance | Rank | Feature   | Importance |
|------|-----------|------------|------|-----------|------------|
| 1    | PAY_0     | 0.2307     | 16   | BILL_AMT5 | 0.0264     |
| 2    | LIMIT_BAL | 0.1036     | 17   | BILL_AMT6 | 0.0260     |
| 3    | BILL_AMT1 | 0.0899     | 18   | PAY_AMT5  | 0.0224     |
| 4    | PAY_2     | 0.0576     | 19   | SEX       | 0.0218     |
| 5    | PAY_3     | 0.0498     |      |           |            |
| 6    | PAY_AMT3  | 0.0473     |      |           |            |
| 7    | PAY_AMT2  | 0.0470     |      |           |            |
| 8    | PAY_AMT1  | 0.0411     |      |           |            |
| 9    | BILL_AMT2 | 0.0403     |      |           |            |
| 10   | PAY_4     | 0.0395     |      |           |            |
| 11   | AGE       | 0.0342     |      |           |            |
| 12   | PAY_AMT6  | 0.0326     |      |           |            |
| 13   | PAY_6     | 0.0314     |      |           |            |
| 14   | PAY_5     | 0.0300     |      |           |            |
| 15   | PAY_AMT4  | 0.0284     |      |           |            |

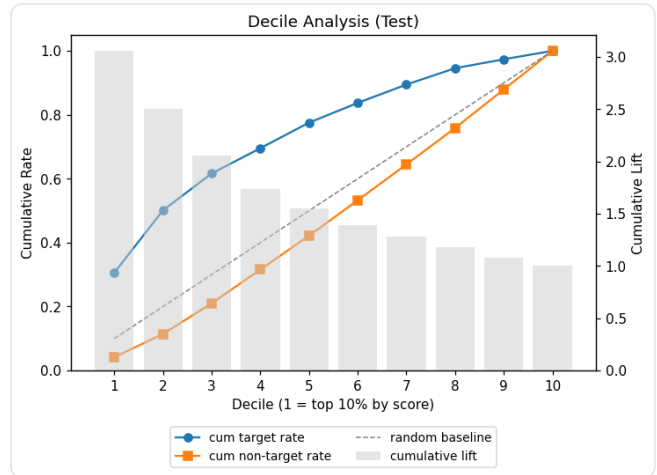
### 3 모델 상세 — ensemble

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



| Decile | Score Range   | Count | Target | Non-Target | Cum Target | Cum NonTarget | Cum Count | Cum %  |
|--------|---------------|-------|--------|------------|------------|---------------|-----------|--------|
| 1      | 0.526 ~ 0.895 | 600   | 406    | 194        | 406        | 194           | 600       | 10.0%  |
| 2      | 0.328 ~ 0.526 | 600   | 259    | 341        | 665        | 535           | 1200      | 20.0%  |
| 3      | 0.213 ~ 0.328 | 600   | 153    | 447        | 818        | 982           | 1800      | 30.0%  |
| 4      | 0.177 ~ 0.213 | 600   | 105    | 495        | 923        | 1477          | 2400      | 40.0%  |
| 5      | 0.153 ~ 0.177 | 600   | 106    | 494        | 1029       | 1971          | 3000      | 50.0%  |
| 6      | 0.132 ~ 0.153 | 600   | 82     | 518        | 1111       | 2489          | 3600      | 60.0%  |
| 7      | 0.114 ~ 0.132 | 600   | 76     | 524        | 1187       | 3013          | 4200      | 70.0%  |
| 8      | 0.099 ~ 0.114 | 600   | 68     | 532        | 1255       | 3545          | 4800      | 80.0%  |
| 9      | 0.079 ~ 0.099 | 600   | 37     | 563        | 1292       | 4108          | 5400      | 90.0%  |
| 10     | 0.021 ~ 0.079 | 600   | 35     | 565        | 1327       | 4673          | 6000      | 100.0% |

Confusion Matrix threshold = 0.5

|        | Pred 0 | Pred 1 |
|--------|--------|--------|
| True 0 | 4437   | 236    |
| True 1 | 872    | 455    |

#### 4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature\_selection.csv)

| Feature   | Selection Frequency    |        | 채택   |
|-----------|------------------------|--------|------|
| LIMIT_BAL | <div><div></div></div> | 100.0% | ✓ 채택 |
| AGE       | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_0     | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_2     | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_3     | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_4     | <div><div></div></div> | 100.0% | ✓ 채택 |
| BILL_AMT1 | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT1  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT2  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT3  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_AMT6  | <div><div></div></div> | 100.0% | ✓ 채택 |
| PAY_5     | <div><div></div></div> | 98.8%  | ✓ 채택 |
| BILL_AMT5 | <div><div></div></div> | 98.8%  | ✓ 채택 |
| PAY_6     | <div><div></div></div> | 97.5%  | ✓ 채택 |
| PAY_AMT4  | <div><div></div></div> | 96.2%  | ✓ 채택 |
| PAY_AMT5  | <div><div></div></div> | 95.0%  | ✓ 채택 |
| SEX       | <div><div></div></div> | 91.2%  | ✓ 채택 |
| BILL_AMT6 | <div><div></div></div> | 88.8%  | ✓ 채택 |
| BILL_AMT2 | <div><div></div></div> | 71.2%  | ✓ 채택 |
| BILL_AMT3 | <div><div></div></div> | 56.2%  | 제외   |
| BILL_AMT4 | <div><div></div></div> | 28.7%  | 제외   |
| EDUCATION | <div><div></div></div> | 21.2%  | 제외   |
| MARRIAGE  | <div><div></div></div> | 15.0%  | 제외   |

## 5 학습 설정 + 튜닝 — ensemble

| 키           | 값                                                                               |
|-------------|---------------------------------------------------------------------------------|
| tuned       | NO                                                                              |
| params (최종) | {'weights': {'elasticnet': 0.4840579071352653, 'catboost': 0.5159420928647347}} |